Mobile Clinical Inbox Prioritization

Product Requirements Document

Author: Yash Bhatt

Note on evidence. This is a portfolio project. I have not spoken to doctors or used real product data. All user needs, figures, and workflows in this document are hypotheses. The prototype and its Admin view use synthetic data.

1. Summary

Providers use the mobile inbox between patients to review lab results, patient messages, refill requests, and administrative items. The list is ordered by time, so an urgent item can sit below routine ones.

This feature suggests an order for the inbox. It shows the reason, the source, and a confidence level for each suggestion. The provider can override it, give feedback on it, or switch back to newest first. A critical lab result also sends a private lock-screen alert. The pilot team watches a monitor with a release gate and an off switch.

The feature does not reply to, close, route, or hide any item.

2. Problem

A provider with a few minutes between appointments needs to find the item that cannot wait. In a time-ordered list, a critical lab result can appear below a scheduling request, and nothing tells the provider which item to open first. Phone alerts today say how many items arrived, not which one matters.

Assumption to verify (H8). On a weak or lost connection, a clinical inbox may show an out-of-date list that looks current. A new critical result could then be missing without the provider knowing. This has not been confirmed.

This document addresses one narrow question: how to suggest an order for urgent items on mobile without hiding items, acting for the provider, or increasing the number of missed urgent items, and how to watch that after launch.

3. Users

These are working assumptions.

- **Provider.** Uses the app between patients, often with one hand, in poor light, and with interruptions. Needs to find urgent items quickly and understand why they were suggested.
- **Nurse or medical assistant.** Sorts routine items first. Needs the same view as the provider.
- **Patient safety lead.** Defines a missed urgent item, reviews the monitor, and can stop the pilot.
- **Product manager.** Owns the monitor, the feedback review, and the release decision with patient safety.

4. Goals and non-goals

Goals

- Help providers reach urgent items sooner.
- Show a plain reason, source, and confidence level for each suggestion.
- Keep the time-ordered list available at all times.
- Alert privately for critical lab results.
- Be honest when the list may be out of date.
- Work for users who need larger text or higher contrast.
- Give the team a way to see problems and switch the feature off.

Non-goals

- Diagnosis or treatment advice.

- Replying to patients, closing items, or routing them.
- Alerts for model-suggested items.
- Learning from a single override or feedback tap.
- Inpatient, emergency, and on-call use.

5. Hypotheses

Each hypothesis must be tested before build.

#	Hypothesis	How to test
H1	Providers check the inbox on mobile in short, interrupted sessions.	Usage data and observation of 2 to 3 providers
H2	Urgent items are sometimes reviewed later than they should be because they sit among routine items.	Time to first review, by item type
H3	Existing flags and filters do not cover urgent patient messages written in free text.	Review of current flags and provider interviews
H4	Providers will use a suggested order only if they can see the reason and return to time order.	Prototype test with 5 to 6 providers
H5	Nurses and medical assistants already sort routine items, so the provider list is partly filtered.	Interviews with 2 nurses or assistants
H6	Patient safety can define "urgent" and "missed" in a way that can be measured.	Working session with the patient safety team
H7	Providers want alerts only for critical lab results, and see alerts for model-suggested items as noise.	Prototype test and alert history review
H8	A meaningful share of provider mobile sessions happen with weak or no signal, and the current app may show an out-of-date list without saying so.	Session data by connection quality, and a test of the current app with the connection off

Stop conditions. Do not proceed if urgent items are already reviewed within the accepted time (H2), if existing rules already solve the problem (H3), or if providers treat the suggestion as an instruction (H4).

6. Scope

First pilot, if hypotheses are confirmed

- Outpatient internal medicine, one practice group.
- Two item types: critical lab results and unread patient messages with symptom language.
- iOS and Android, English only.
- Alerts for critical lab flags only.
- Snooze, feedback, weak-signal behavior, and accessibility settings.
- The pilot monitor and off switch.

Deferred

- Drafted replies and automatic escalation.
- Ranking of billing and administrative items.
- Learning from individual provider behavior or feedback.
- Alerts for model-suggested items.
- Inpatient, emergency, and on-call use.

7. User flow

1. The provider receives a private alert for a critical lab result, or opens the inbox directly.

2. The inbox opens. A banner explains that the order is a suggestion.
3. Urgent items are labeled and shown first. All other items stay on the list.
4. The provider opens an item and reads the reasons, source, and confidence level, and can say whether the reasons were right.
5. The provider chooses "I'll review now", "Not urgent", or "Remind me later". None sends a message or closes the item.
6. The provider can switch to time order, or change text size and contrast, in settings at any point.
7. With no signal, the inbox says when it was last updated, pauses suggestions, and warns that new critical results may be missing.

The clickable prototype shows each step, plus the lock screen, weak-signal state, outage state, and the Admin view.

8. Requirements

ID	Requirement	Priority
R1	The time-ordered inbox is always available and is used if suggestions fail or are turned off.	Must
R2	Suggested order never hides or removes any item.	Must
R3	Each suggested view states that the order is a suggestion, not a diagnosis.	Must
R4	Each suggested item shows two or three plain reasons, the source (rule or model), and a confidence level in words. A score alone is not shown.	Must
R5	"Not urgent" changes the display order only. It does not close, reply to, or route the item.	Must
R6	"I'll review now" records attention only. Reply and close stay in the existing controls.	Must
R7	Every suggestion, view, feedback answer, snooze, and override is saved in an audit log with user, item, time, and model version.	Must
R8	A critical lab flag cannot be ranked lower than the existing rule would rank it.	Must
R9	The pilot team and patient safety can see missed-urgent and override rates.	Must
R10	The provider's choice of time order is remembered.	Should
R11	Nurses and providers see the same item status.	Should
R12	Send a lock-screen alert only for critical lab flags. The alert shows no patient name or result value. Tapping it opens that item after unlock. Model-suggested items never send an alert. Alerts do not depend on the model.	Must
R13	On a suggested item, the provider can mark the reasons right or wrong, and give a short reason. Feedback is logged and reviewed by a person. It does not retrain the model on its own.	Should
R14	The provider can snooze an item to a chosen time. A snoozed item stays in the inbox under its own heading and returns at that time. Critical lab results cannot be snoozed.	Should
R15	With weak or no signal, show the last synced list in newest-first order with the time it was last updated, and pause suggestions. Warn that new critical results may be missing until the connection returns. Reading works. Actions that must be recorded wait for a connection, and the app says so in plain words instead of a generic error.	Must
R16	Support larger text and high contrast. Touch targets are at least 44 points. Reasons and confidence are readable by a screen reader.	Must
R17	A pilot monitor shows the release gate, review time, missed urgent by item type, usage, and fairness and data checks. Named owners can turn suggestions off for all pilot users without an app release.	Must

9. Rules for AI behavior

- Use only information the provider can already see in the chart.
- Output only a rank, short reasons, a source, and a confidence level.
- Never diagnose, draft replies, close items, or label an item "safe to ignore".
- A person takes every clinical action.
- The critical lab rule works without the model. The model may add items, but may not lower a rule-based critical item.
- A new model version needs patient safety review before it reaches providers.

10. Acceptance criteria

1. **No hiding.** Given a critical lab below a newer scheduling item, when suggested order loads, the lab appears first, the scheduling item stays visible, and time order is one tap away.
2. **Reason shown.** When a provider opens a suggested item, at least two reasons, the source, and a confidence level are visible before any action.
3. **Override contained.** When the provider taps "Not urgent", the order updates and the action is logged. No message is sent, no item is closed, and no model training starts.
4. **Safe fallback.** If the suggestion service fails, the inbox shows time order and says suggestions are unavailable.
5. **Audit.** Every suggestion and action is logged. Entries cannot be edited.
6. **Private alert.** Given a critical lab flag, when the phone is locked, the alert shows no name and no value, and no alert is sent for a model-suggested item.
7. **Snooze is safe.** Given a critical lab result, the snooze control is unavailable. Given any other item, a snoozed item is visible under Snoozed and returns at the chosen time.
8. **Weak signal.** Given no connection, the inbox shows the last synced list in newest-first order with the time it was last updated, suggestions are paused, and a notice warns that new critical results may be missing. Reading works. Reply and close show a plain "no signal" message. Actions that must be recorded are unavailable until reconnection.
9. **Off switch.** Given an owner turns suggestions off, providers see newest first and a notice, and critical lab alerts continue.
10. **Feedback.** Given the provider marks reasons wrong, the answer and reason are logged, shown in the monitor, and do not change the model.

11. Success measures

Primary measure. Median time from an urgent item arriving to its first review, on mobile. No target is set until a baseline is measured.

Safety measure (release gate). Share of urgent items not reviewed within the time set by patient safety. If this rises, the feature does not expand, even if review time improves. The monitor shows this beside review time so speed cannot pass alone.

Supporting measures

- Share of sessions where the provider opens a suggested item first.
- Share of suggestions overridden.
- Share of sessions switched back to time order.
- Share of sessions that open the reason.
- Share of feedback answers that mark the reasons right.
- Missed urgent rate by item type, hour of day, and practice size.

12. Risks

Risk	Impact	Response
Providers treat the suggestion as an order	They stop checking the full list	Clear banner, no automatic actions, test in H4
A critical lab is ranked too low	Delayed care	Requirement R8 and the missed-urgent gate
"Urgent" is not clearly defined	Results cannot be trusted	Patient safety sets the definition first (H6)
One override or feedback tap changes the model	Unstable behavior	No learning from single taps
Existing rules already solve the problem	The feature adds little	Check rule coverage (H3)
Lock-screen alert reveals patient information	Privacy harm	No name or value on the lock screen (R12)
Alerts for too many items	Providers ignore all alerts	Critical lab flags only (H7)
An out-of-date list looks current on weak signal	A new critical result is missed without the provider knowing	Show the last update time, pause suggestions, and warn about missing critical results (R15)
Snooze delays a critical item	Delayed care	Critical lab results cannot be snoozed (R14)
Time improves but missed urgent rises	False sense of success	Monitor shows both, and the gate decides (R17)

13. Dependencies

- Patient safety approval of the urgent definition, review time, and alert wording.
- Access to inbox event data, or agreement to add tracking first.
- A data science owner for model review and feedback review.
- UX support for mobile design and accessibility.
- Security and compliance approval of the audit log and alert content.
- Mobile push service and OS support for private lock-screen content.
- Event data by connection quality, to test H8.
- A clinical sponsor in the pilot group.

14. Rollout

1. **Baseline.** Measure current review times. Agree on the urgent definition. Check how the current app behaves with no signal.
2. **Design and safety review.** Test the prototype with providers. Patient safety approves intended use and alert content.
3. **Offline check.** Test suggestions on past data and review every miss.
4. **Silent run.** Calculate suggestions without showing them. Compare with what providers opened first.
5. **Pilot.** One practice group, with the monitor live and the off switch tested before the first provider sees a suggestion. Review weekly.
6. **Expansion decision.** Expand only if missed-urgent items do not increase.